BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Project Review and Assessment Feedback

Name _____ **Class** _____ **Date** _____ **Time** _____ **Teacher** _____

Do Now - Four retrieval questions

1. What is the purpose of testing a program against expected outputs?

2. What can unsafe SQL input let an attacker change?
   A) The CPU's clock speed
   B) An optical disc's capacity
   C) A database query
   D) A cable's length

3. Can an array be passed to or returned from a subprogram?

4. A school updates its anti-malware. "Anti-malware can detect every future threat." Correct this. Why are other controls needed?

Today's objectives

Improve programs using test results and feedback. Apply this to another software scenario.

Key vocabulary - Read, cover, recall

Test evidence
A table should record test input, expected result, actual result and outcome.

Fault diagnosis
Compare expectation and observed behaviour to identify a specific error.

Refinement
Change the faulty instruction and run the affected test again.

Feedback
Use genuine tester or teacher comments to choose improvements, not guessed marks.

BLACK PEN: Write one definition in your own words.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Learn and practise

New Learning - Key ideas

Test evidence
A table should record test input, expected result, actual result and outcome. Consider this in a project correction.

Fault diagnosis
Compare expectation and observed behaviour to identify a specific error. Consider this in a quiz total review.

Refinement
Change the faulty instruction and run the affected test again. Consider this in a ticket app test failure.

Feedback
Use genuine tester or teacher comments to choose improvements, not guessed marks. Consider this in a club database feedback.

I Do - Worked model

A project correction needs test evidence and fault diagnosis. Explain why. [4 marks]

- Identify the inputs, constraints and expected result.
- Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1].
- Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1].
- Give a specific point and a linked reason for each feature.

We Do - Guided practice

1. In a quiz total review, explain the purpose of test evidence. [2 marks]

2. A ticket app test failure needs fault diagnosis. Explain its role. [2 marks]

3. A club database feedback uses refinement and feedback. Explain both. [4 marks]

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Think and write

Code First - Predict > Write > Test > Improve

YOUR TASK: Correct the calculation so it prints the price of three £6 items.

```
unitPrice = 6
quantity = 3
total = unitPrice + quantity
print(total)
```

You Do - Independent exam questions

Q2. State TWO concepts relevant to this topic. [2 marks]

Q3. Explain how fault diagnosis and refinement help in a café receipt bug. [4 marks]

Q4. Explain the roles of test evidence, refinement and feedback in a pupil scoring app. [6 marks]

Exit Ticket - True or False?

A table should record test input, expected result, actual result and outcome.

TRUE [ ] **FALSE [ ]**

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick correct answers; rewrite mistakes.

Review & improve

GREEN PEN: Tick each correct point; correct mistakes.

Do Now - Model answers

1. To compare its actual behaviour with the requirements and identify faults.

2. C - A database query

3. Yes. A subprogram can receive an array; a function can return an array.

4. Anti-malware detects, quarantines or removes malware. Updates help it recognise threats. It cannot stop every attack, so other controls are needed.

I Do - Explain the model

Identify the inputs, constraints and expected result.

Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1].

Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1].

Give a specific point and a linked reason for each feature.

We Do - Model answers

1. Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1].

2. Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1].

3. Refinement [1]; Change the faulty instruction and run the affected test again. [1]. Feedback [1]; Use genuine tester or teacher comments to choose improvements, not guessed marks. [1].

Code First - Answer

```
unitPrice = 6
quantity = 3
total = unitPrice * quantity
print(total)
```

Expected output: 18

You Do - Self-mark

Q2: Test evidence [1]; Refinement [1].

Q3: Fault diagnosis [1]; Compare expectation and observed behaviour to identify a specific error. [1]. Refinement [1]; Change the faulty instruction and run the affected test again. [1].

Q4: Test evidence [1]; A table should record test input, expected result, actual result and outcome. [1]. Refinement [1]; Change the faulty instruction and run the affected test again. [1]. Feedback [1]; Use genuine tester or teacher comments to choose improvements, not guessed marks. [1].

GREEN PEN: Choose the hardest answer. Correct it on lined paper.